

数据库系统原理

— Course Introduction

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DBS 现状与分析



数据库作为基础软件的“三驾马车”之一，是许多应用与系统的结实底座，然而，据《2021-2022中国基础软件开发者调查报告》显示，在从事与数据库相关工作的开发者中，53%为业务开发，21%为数据分析师，另有DBA占比19%，只有7%的开发者懂内核研发，可见数据库内核人才之匮乏。

Examples

淘宝 / 天猫如何应对双十一海量高并发访问？

12306 系统架构、抢票机制？

银行系统如何通过数据备份、双活、冗余机制保证数据高可用

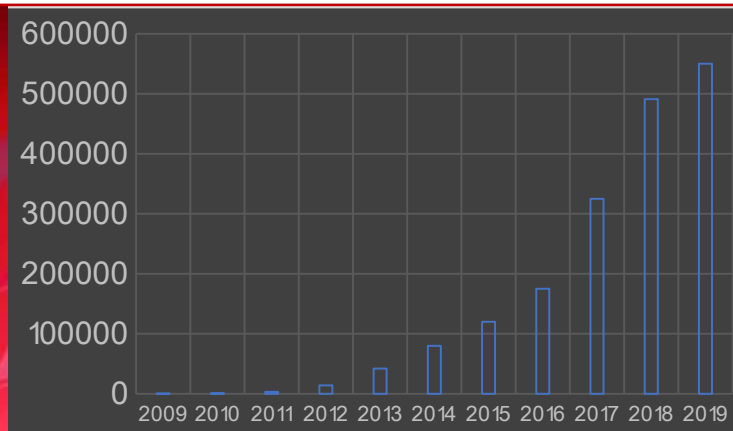
如何设计一个秒杀系统？

美团系统架构？

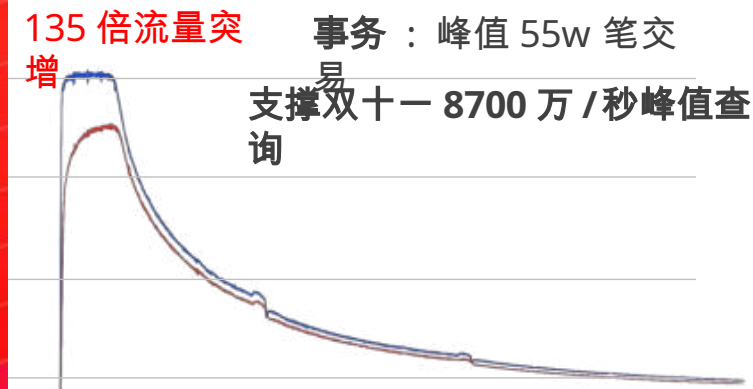
OceanBase 数据库

openGauss 数据库

数据仓库，数据中台，数据湖
知识图谱；
云原生数据库



天猫双十一历年峰值交易笔数



2019 双十一零点数据库请求曲线图

We use three levels of mastery, defined as:

- ***Familiarity***: The student understands what a concept is or what it means. This level of mastery concerns a basic awareness of a concept as opposed to expecting real facility with its application. It provides an answer to the question “What do you know about this?” 是什么
- ***Usage***: The student is able to use or apply a concept in a concrete way. Using a concept may include, for example, appropriately using a specific concept in a program, using a particular proof technique, or performing a particular analysis. It provides an answer to the question “What do you know how to do?” 如何做
- ***Assessment***: The student is able to consider a concept from multiple viewpoints and/or justify the selection of a particular approach to solve a problem. This level of mastery implies more than using a concept; it involves the ability to select an appropriate approach from understood alternatives. It provides an answer to the question “Why would you do that?” 为什么

- SQL
- 数据库设计
- 索引、触发器、游标
- JDBC、ODBC



- 存储引擎
 - 页面设计、WAL、HA
 - Buffer、redo/undo
- 事务处理
 - OCC、2PL、MVCC、2PC
- 查询优化
 - 重写、代价、计划、执行

- 数据模型、ER模型
- 关系代数
- 数据库范式
- SQL查询优化器
- 数据库事务处理

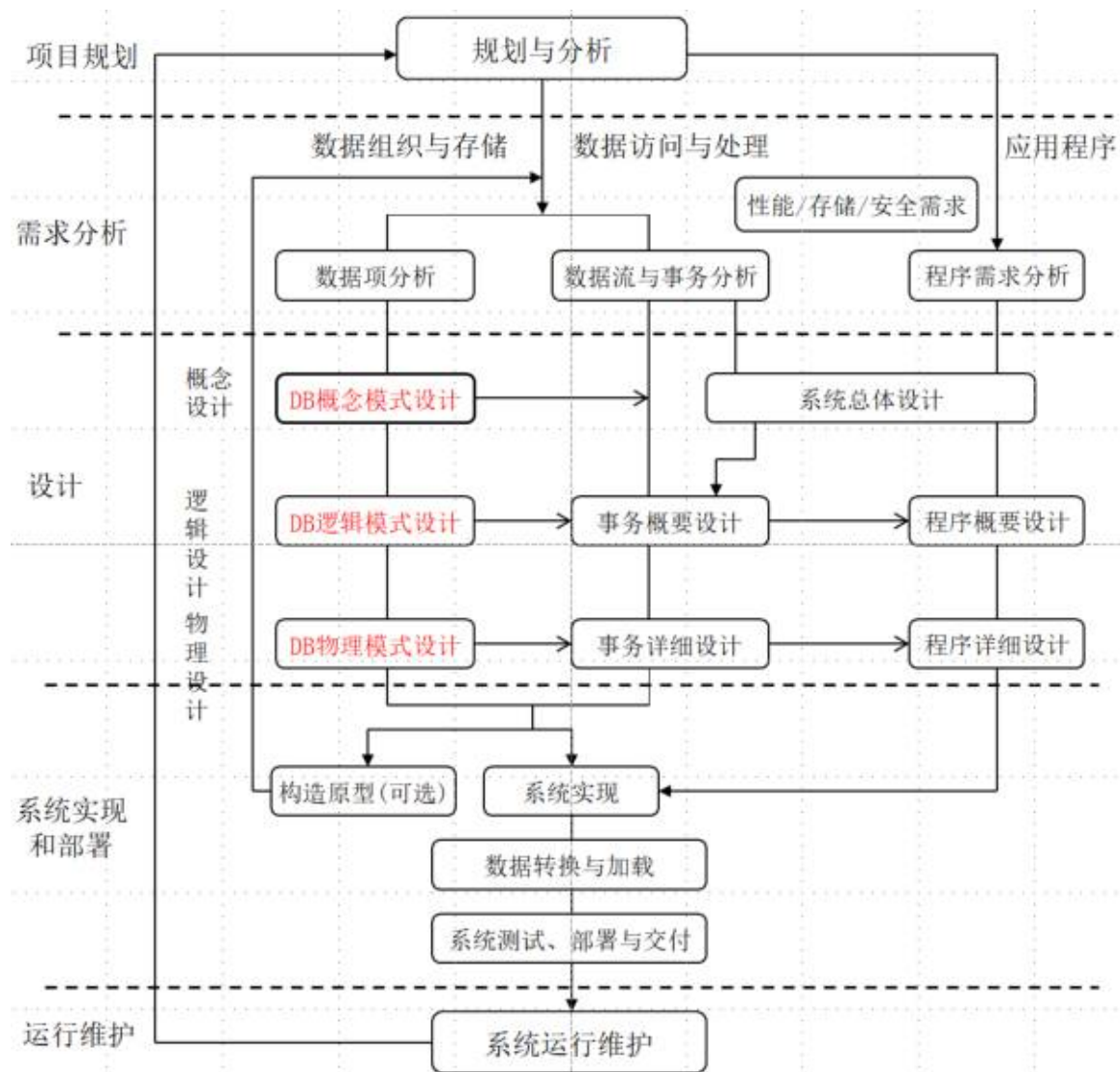
- OLAP：行存、向量化
- HTAP：行列存
- 云数据库
- NOSQL
- NEWSQL
- Multi-Model

数据库 / 数据仓
库 / 数据中台 / 数据
湖，

云原生数据库

- 1. 引论【第1章】
 - 数据库系统概念、组成，体系结构，数据库语言，数据模型
- 2. 关系数据库【第2~6章】
 - 关系数据模型，SQL语言，关系代数
- 3. 数据库设计【第7~8章】
 - 实体-联系模型，关系规范化
- 4. 数据存储【第10~11章】
 - 数据文件组织与访问，索引
- 5. 查询处理及优化【第12~13章】
 - 查询执行计划及成本估算，启发式查询优化
- 6. 事务管理【第14~16章】
 - 事务，事务调度，隔离级别，并发控制，恢复

数据库应用系统 (DBAS) 全生命周期模型



- Textbook

- Database System Concepts (第 7 版 , 影印版) 高教出版社 , 2021 年 3 月
- [Database System Concepts - 7th edition \(os-book.com\)](https://www.db-book.com/db7/index.html) ,
<https://www.db-book.com/db7/index.html>
【原版电子书】
- Bilingual teaching and English textbooks
- Bilingual teaching in universities is an important policy decided by Ministry of Education, also by BUPT
- it is better for students to read the English textbook written by Abraham Silberschat et al. directly, not to use Chinese-version of the textbook for this course

- Mid-term test, participant, exercises, experiments, final examination
 - Grading : 10% + 10%+ 10% + 10% + 60%
- 答疑
 - 周三下午 14:00—18:00 ，明光楼 312 ；
 - 研究生助教：



Thanks for your attention

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home and
organization for a fully connected,
intelligent world.

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